

Question and Answer

Artificial Intelligence and Machine Learning

(6CS012)

Rahul Dev Banjara

L6CG1 | University ID: 2407061

Tutor: Ms. Sunita Parajuli

Module Leader: Mr. Siman Giri

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Part I : Question and Answer

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1 Long Question Answer

Selected Question: Question 1

You are a Data Scientist at a leading digital payment platform such as eSewa, working on large-scale user transaction data.

Working with transaction data at the scale of a platform like eSewa is both exciting and challenging. Assuming, I am a data scientist at eSewa, that means we have millions of transactions everyday but very less labeled data, because manually labeling fraud or risky behavior takes lot of time, money and human effort. This is exactly where unsupervised learning becomes very valuable. Below I have described two high-impact business problems where unsupervised learning can provide real business value.

Problem 1: Detecting Unofficial Agent Accounts

Real-World Context: People in several villages of Nepal, are unaware of how to use eSewa app. There is always one clever person who is an unofficial agent. They collect cash from villagers and pay their electricity bills/Top-ups from their personal eSewa account to earn Reward Points or small cash commissions. eSewa wants these users to be official Agents for better monitoring, but we don't know who they are to proceed further to making them. Labeled data is not available here since, from a system point of view, they are just a very active normal user.

Proposed Approach: Clustering on Behavioral Density using K-Means Clustering. We are considering two simple features here: Number of unique consumer IDS (for eg. 50 different NEA electricity meter numbers paid from one account) and Frequency of Transactions per month. The normal user pays 1-2 bills. This unofficial agent pays 40 bills a month but from a personal profile. These users are very distinct from the average user behavior, and will seem to be in a completely separate cluster when separated by K-Means.

System Integration: This is a batch job that is performed monthly. The output is taken as the High-Utility Cluster. The Sales team then receives a pure list of these users and calls them to provide an Official Agent Partnership. What this is is a very clever business strategy it's basically making what was a leakage and informal deals, into a formal and regulated business channel. eSewa ensures greater compliance coverage and the user receives the correct agent resources and commissions.

Limitation: In large families it is common for one person in Nepal to pay for bills for the whole extended family, Aama, Kaka, Mama and close relatives as well. The model may be confused as to what a "Helpful Son" is and what an "Unofficial Agent" is.

Problem 2: User Behavior Analysis for Credit Scoring

Real-World Context: eSewa is working towards providing digital micro-loans and financial products. The problem is that a very large percentage of users, particularly those of semi-urban and rural background, don't have any formal credit history. These users are not assessable by the traditional banks. However, eSewa has a lot of behavioral data to analyze: Remittance payments, wallet top-up frequency, airline bookings, QR merchant transactions, and payments of their utility bills. We can use this to gain insights into the financial reliability of a user without relying on labelled default data.

Proposed Approach: I would use DBSCAN for segmentation of customers. DBSCAN doesn't require us to define the number of clusters earlier and naturally deals with outliers as noised. This is crucial in this case because some users' behavior will be very irregular or atypical that we shouldn't be combining with any stable segment. Users can be grouped by features such as spending consistency, transaction frequency, average ticket size and diversity of merchant categories.

System Integration: The Risk team can pre-approve loan limits for users who are in the financially stable and high consistency clusters in batch mode using the cluster outputs. In real time, the cluster label can be used to create customized rewards, such as a user who is a regular utility payer and consistent spender can be pre-approved for a smaller credit line within the eSewa app without having to fill out any paper documents.

Limitation: The clusters may be very wide apart and difficult to understand due to the hundreds of features included in the transaction data. The more dimensions there are, the less reliable distance-based algorithms, such as DBSCAN, will be. Dimensionality reduction techniques such as PCA would have to first be applied before clustering to obtain meaningful and stable results. This will complicate the pipeline, and it can also make it somewhat less easy to describe clusters to business stakeholders simply.

2 Short Question Answers

2.1 Section 1: Overfitting - Question 2

Describe at least two techniques used to reduce overfitting.

In deep learning, especially when the model is large compared to the number of training data, overfitting is a common issue. The model just learns patterns of the training set rather than learning generalizable features, and as a result it performs good on training data but poor on new data.

Technique 1: Dropout

The intuition behind dropout is to randomly zero out a fraction of the neuron activations during every forward pass of the training process. The idea is that this will not allow the network to become too reliant on a specific group of neurons. The model has to learn more distributed and robust representations since each training step involves different subsets of the network. During inference, all of the neurons are turned on and their output intensities are appropriately scaled. Dropout is found to be helpful for image classification in practice. For instance, in CNN models trained on a medical imaging dataset, adding dropout layers after fully connected layers can minimize the difference between the training and validation accuracy, a direct measure of overfitting.

Technique 2: Early Stopping

Early stopping is a technique in which a validation measurement (typically validation loss) is watched during training and training stops once it has stopped to improve, even though the training loss may continue to decrease. As the name suggests, the idea is straightforward: once a model is good enough to learn the noise in the training data, more training makes it worse on the test data. Typically, classification problems, such as sentiment analysis, see training loss keep on decreasing but val loss begin to increase after a certain epoch. Early stopping catches that point and saves model weights at the best validation epoch. The main benefit is that it doesn't need to change the architecture of the model, and it's quite simple to implement.

Both methods address overfitting problems in different ways. Dropout adds regularization during training through stochasticity, and early stopping regulates the amount of time the model is given to fit the training data. When used together they are more effective than either approach alone.

2.2 Section 2: Neural Network Architecture - Question 2

Compare a standard feedforward neural network (FCN) with an autoencoder.

A feedforward neural network (also known as a fully connected network (FCN)) is a standard neural network that receives an input vector and output a label or value. All layers are connected with each other and the objective is to minimize the difference between the predicted output and the ground truth label. The input and output dimensions typically aren't equal; that is, when using a digit classification example, a 784-dimensional image is given and the digit class is 10-dimensional, represented by a softmax output.

The goal of an autoencoder is quite different. It's an unsupervised architecture that attempts to recreate its own input data at its output. The architecture consists of two components: one is an encoder which is a mapping from the input space into a lower dimensional space (also known as the latent space or bottleneck), and the second is a decoder which is a mapping from the lower dimensional space (latent space or bottleneck) back into the input space. The objective of this architecture is to recreate its own input data at its output. One important difference is that there are equal number of input and output dimensions, with the bottleneck in the middle knowingly made smaller than either.

A Simple Example

Say we have 28x28 grayscale images (784 pixels). A standard FCN might take this as input and output a class label (it should be there in the dataset). An autoencoder would take the same 784-dimensional input, compress it to, say, 32 dimensions in the bottleneck, and then try to reconstruct the original 784-dimensional image at the output. The model is trained by minimizing reconstruction loss, not classification loss.

For an example, we have 28x28 grayscale images (784 pixels). This could be used as an input and output of a standard FCN. Autoencoder would compress it down to, say 32 dimensions, in the bottleneck layer, and then attempt to reconstruct the original image of 784 dimensions at the output. The training criterion for the model is not classification loss, but reconstruction loss.

Application: Anomaly Detection

Autoencoders are known to be used for anomaly detection. The training of the autoencoder is supposed to be limited to normal data. The model will have low reconstruction error in the normal cases and high reconstruction error in the anomalous cases, because the model is trained to reconstruct normal patterns. For instance, in a network security environment, an autoencoder, whose input is a normal traffic log, will give high reconstruction error if it receives abnormal traffic patterns, which are considered possible intrusions. The model identifies what

is “normal” and considers anything outside of that normal as an “outlier” which can be helpful when there is a lack or limited availability of labeled anomaly data.